A mergeable measure of sentiment for text of any length

Ayaz Khan
Department of Statistics, Columbia University, New York, NY, USA. Code and data: https://github.com/ayazkhan27/anonymity

Abstract pending the last experiments.

Sentiment analysis turns a piece of writing into a number that says how positive or negative it is. The number is computed for product reviews, posts, news, clinical notes and parliamentary speech, often for millions of texts at a time [1–5]. Two families of method are in use. Lexicon methods look each word up in a list of human word ratings and combine the values by rule [6–9]. Classifiers, now mostly fine-tuned transformers, are trained on labelled text and return a class probability [10–12].

Both families return a bounded number for each text: a probability, a mean, or a sum squashed into the interval from −1 to 1. A bounded score is intensive in the sense that temperature is intensive. It does not grow with the amount of text, and the value for a whole cannot be computed from the values of its parts. Three practical problems follow. A transformer reads a few hundred words at a time [13], so a long document is cut into chunks whose scores are averaged, and the average depends on where the cuts fall. A squashed sum saturates: in the novels studied below, VADER's compound score is beyond ±0.99 for 87% of whole chapters. And an average of averages can rank two bodies of text in the opposite order to their pooled values, the reversal known from Simpson's paradox [14–16].

Here I treat sentiment as a quantity that adds. Every word carries a positive and a negative charge taken from human word ratings. A span of text is summarised by a short vector, its signature, which holds counts and sums of charge weighted by powers of word position. Signatures of adjacent spans merge exactly into the signature of the joined span, so a novel is scored in one pass and every chapter, page and sentence can be read off afterwards without scoring anything again. Scores that people can compare, such as net charge per thousand words or a predicted human rating with an error bar, are read from the signature at whatever scale is wanted.

The parts have precedents. The hedonometer averages human word ratings [2,6]. Word-count tools and studies of mood keep positive and negative words apart [3,9,17], in line with evidence that the two are separable in evaluation [18]. Political scientists score texts from counts of opposed items on a log-ratio scale with a stated uncertainty [4,19]. Summaries that merge exactly across partitions, including sums weighted by polynomial or exponential functions of time, are standard in stream processing [20–23]. Emotional arcs of stories have been drawn with moving windows, transforms and fitted polynomials [24–26]. This paper joins these into one object for sentiment, states which position weights such an object can carry, and tests it against human ratings.

The work began with a manuscript I wrote in 2024, which proposed that feeling in text could be scored as a number from word values, a decay over position and a penalty for words that carry no feeling. It made two claims about how people write and read. Front-loading: words that carry feeling come early in a text and count for more there. Dilution: words that carry none weaken the feeling of the whole. I wrote both down as hypotheses with decision rules, and committed them to the project repository, before scoring any evaluation data (Methods). Front-loading fails in every corpus examined. Charged words sit slightly later than the midpoint of a text, and in 154,457 human-rated phrase pairs the later part carries more weight than the earlier one. Dilution holds, with a twist that the 2024 manuscript did not anticipate: neutral words weaken a rating when they follow the feeling and leave it almost unchanged when they come before it.

Results

A signature that merges

Take a text as a sequence of words. Word i has a positive charge h_i and a negative charge d_i, each between 0 and 1, at most one of them nonzero. The values come from two open sets of human word ratings, the labMT list [2] (10,222 words, 50 raters each) and the VADER list [7] (7,494 entries, 10 raters each). On the 1,366 words they share, the two sets correlate at $r = 0.95$. I put both on one scale, on which 1 is the positive end of the rating scale and 0 its neutral point, and average them (Methods). Positive charge is measured in hedons (hd) and negative charge in dolors (dl), the names that moral philosophy uses for units of pleasure and pain [27,28]. One hedon is the charge of a word that every rater places at the top of the scale; *love* carries 0.85 hd and *terrible* 0.55 dl. A word rated within a quarter of the scale from neutral gets no charge and is called a buffer word, following the neutral band of the hedonometer [2]. In the six rated corpora used here, 23% of running words are charged.

The signature of a span holds the number of words L, the number of charged words E, and the sums

$$S_k^+ = \sum_i h_i\, i^k, \qquad S_k^- = \sum_i d_i\, i^k, \qquad k = 0, \ldots, 3, \tag{1}$$

where i is the position of the word in the span, counted from zero. S_0^+ and S_0^- are the span's total hedons and dolors. The signature also holds the hedons and dolors that fall under a negation, after an intensifier, after a diminisher, and after a contrast word such as *but*, together with the sum of squared charges (Methods). That is 19 numbers, whatever the length of the span (Fig. 1a,b).

Signatures merge. If span A is followed by span B, the counts and totals add, and the position sums of B are shifted by the length of A:

$$S_k(AB) = S_k(A) + \sum_{j=0}^{k} \binom{k}{j} L_A^{\,k-j}\, S_j(B). \tag{2}$$

The operation is associative and can be undone, so the signature of B follows from those of AB and A (Proposition 1, Methods). A text of any length is therefore scored in one pass with constant memory. Parts can be scored separately and merged in reading order to the same result as a single pass. A table of prefix signatures, one row per sentence, gives the signature of any contiguous run of sentences from two of its rows. In integer arithmetic these statements hold to the last digit, and the reference implementation, which counts in thousandths of a hedon, is tested for exact equality on random texts. The fast implementation uses double-precision floats; over 356 novels its merged and single-pass signatures agree to a relative error below 1×10^{-13}.

Which position weights allow a merge? Suppose a summary keeps finitely many sums $\sum_i c_i\, \varphi(i)$ of charge against weight functions φ. The summary of AB can be computed from the summaries of A and B and their lengths exactly when the weight functions span a space that is closed under shifting the position, and such weights are

combinations of $i^k \rho^i$: polynomials, exponentials and their products (Proposition 2, Methods). The statement is an instance of a classical fact about translation-invariant function spaces[29,30] and about functions on strings with finite-dimensional representations[31,32], and stream-processing systems rely on it[20]. For sentiment it says two things. The decay factor of the 2024 manuscript, a weight that falls geometrically with position, is one of the few position weights that survive a merge, at the cost of one number per decay rate. And among mergeable weights, polynomials alone can be re-expressed in relative position, that is, position divided by a length that is unknown until the text ends. The signature keeps polynomial sums for that reason: the sum of charge under any cubic function of relative position can be read from it, for a post and for a novel alike.

Reading a score from a signature

A signature is extensive. Scores that can be compared between texts are read from it (Fig. 1c).

Totals are the hedons $H = S_0^+$ and dolors $D = S_0^-$ over L words. *Density* is the net charge per thousand words, $1000\,(H - D)/L$; the hedonometer's average happiness is a density taken over charged words only.

Level follows a model from the psychology of judgement. In Anderson's averaging model a judgement is a weighted mean of the pieces of information received and an initial neutral impression[33], and information that says nothing pulls a judgement toward neutral, the dilution effect[34]. For words this gives

$$\text{level} = 1000\,\frac{H - D}{a + E + \kappa B}, \tag{3}$$

where $B = L - E$ counts buffer words, a is the weight of the neutral first impression in units of charged words, and κ is the weight of a buffer word relative to a charged one. With $\kappa = 0$ buffer words do nothing; with $a = 0$ and $\kappa = 1$ the level is the density. The denominator is the exposure, the amount of evidence behind the score, and it gives the level a standard error that shrinks as a text grows (Methods).

Calibrated valence is a predicted human rating. It is a linear combination of ten signature entries, each divided by the exposure: hedons, dolors, three position moments of net charge, and the charge under each operator. Given a and κ, the coefficients solve a ridge regression against human ratings, a convex problem with a closed-form solution; a and κ are chosen on a grid (Methods). The fitted coefficients say how much a negation flips a word and how position is weighed; they are estimates with intervals, in the tradition of rule constants measured from ratings[7,35].

The *arc* is the least-squares cubic fit of net charge against relative position. It has a closed form in the position sums (Methods), so it costs nothing beyond the signature.

Agreement with human ratings

I compared scores with human ratings on six corpora: sentences from six genres (EmoBank[36]), tweets (SemEval-2018[37]), Facebook posts[38], film-review sentences (the Stanford Sentiment Treebank, SST[39]), whole film reviews with the critic's rating[40], and Yelp reviews with star ratings[41]. The first five carry continuous ratings and form the primary set. Every constant was fitted leave-one-corpus-out: for each target corpus, a, κ and the ten coefficients came from the other five, and scores were then computed on the target's held-out test portion (Methods). Each of the six fits chose $a = 2/4$ and $\kappa = 0.2/0.5$.

Averaged over the five primary corpora, the calibrated score correlates with human ratings at $r = 0.51$, against 0.49 for VADER[7], 0.44 for TextBlob and 0.42 for the hedonometer[2] (Fig. 2a; all values with intervals in Extended Data Table 2). The paired difference from VADER excludes zero in favour of the signature in 3 of six corpora and in favour of VADER in 0. The level of equation (3), which has no fitted coefficients, reaches 0.48. This meets a criterion that an earlier version of the method failed (Methods), by a small margin.

Methods trained on labelled text do better. A bag-of-words ridge regression trained on each corpus's own training portion averages 0.61. DistilBERT fine-tuned on SST-2[12] averages 0.59 and a RoBERTa model fine-tuned on tweets[42] 0.68. The signature is not a replacement for these where accuracy on one text matters most.

Removing parts of the signature shows what earns its place (Fig. 2c). Without the operator channels the mean correlation falls by 0.034; without dilution ($\kappa = 0$) by 0.015; without the neutral prior ($a = 0$) by 0.010. Merging hedons and dolors into one channel costs 0.002 and dropping the position moments 0.000: neither matters for agreement with raters, though both are needed for the readouts below.

The fitted coefficients have plain readings (shipped model, fitted on all six corpora; bootstrap intervals). A dolor moves the predicted rating 1.63 [1.54, 1.73] times as far as a hedon. A negated positive word counts as -1.60 [-1.85, -1.37] times its charge, about what the same amount of negative charge would do, while a negated negative word counts as -0.01 [-0.23, +0.20]: *not good* is bad, and *not bad* is neutral. An intensifier adds +1.02 [+0.85, +1.19] times the charge of the word it modifies. The data do not settle the effect of diminishers (-0.33 [-0.91, +0.09]) or of contrast words (-0.06 [-0.14, +0.02]); in particular they give no support to the rule that the clause after *but* counts for more[7].

Word charges can also be fitted. With the exposure and the operator constants held fixed, the predicted rating is linear in the vector of charges, so the charges that best predict ratings, pulled toward the human word norms by a ridge penalty, solve one convex problem (Methods). Fitted leave-one-corpus-out, these refined charges raise the mean correlation to 0.55, the same as a bag-of-words ridge trained on the same five corpora (0.55). The gain is uneven. It is large on review text (SST 0.44 to 0.55, film reviews 0.29 to 0.49) and negative on the mixed genres of EmoBank (0.48 to 0.41) and on Facebook posts (0.62 to 0.58). Charges learned from ratings carry the genre of the texts they were learned from; human word norms travel better.

One hoped-for property did not hold. The standard error of a level shrinks with exposure, but the error of the calibrated score against human raters does not: in none of the six corpora did squared error fall as exposure rose (Extended Data Fig. 7a). The standard error says how well a text's own words pin down its level. It is not a prediction interval for a human rating.

Cost

Throughput benchmark pending.

Where the feeling sits

Front-loading makes three predictions, and I tested each (Fig. 3).

The first is that charged words come early. They do not. The charge-weighted mean position, on a scale from −1 at the first word to +1 at the last, is above zero in all 8 kinds of text examined, from +0.002 in Yelp reviews to +0.105 in tweets, with every interval excluding zero (Extended Data Table 3). In EmoBank sentences, SST sentences and tweets the first tenth of the words carries 0.65 to 0.75 of the text's mean charge per word, and in those three and in Facebook posts the last tenth carries 1.34 to 1.55 (Fig. 3a). English sentences tend to open with determiners and pronouns and close on content. Facebook posts open at the mean (1.02). Yelp reviews are the one place with a charged opening, at 1.09 times the mean, matched by the close at 1.11. Negative charge sits later than positive charge in 7 of the 8.

The second prediction is that readers weigh early words more. The

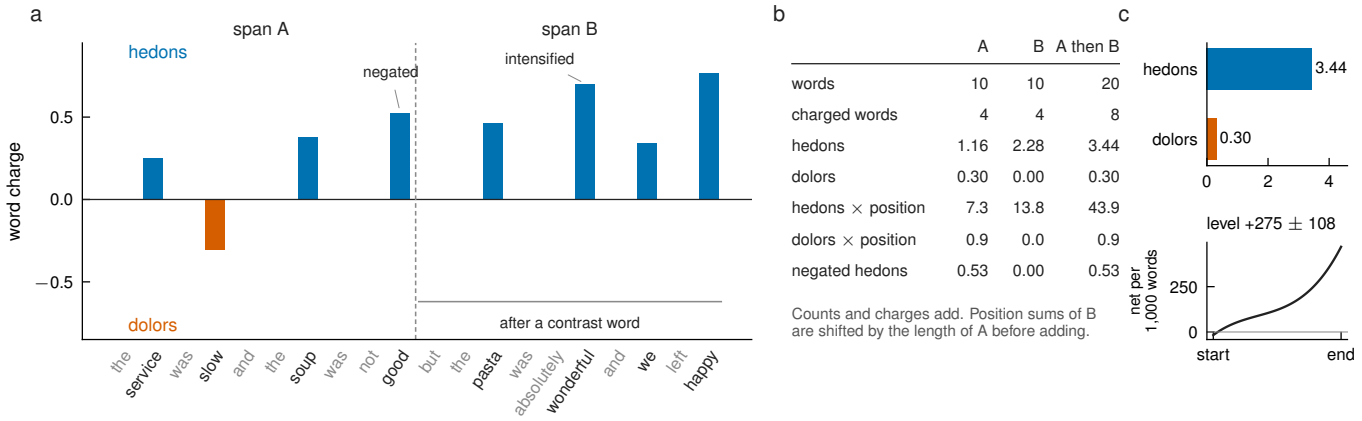


Figure 1 Charge, signature, merge, readouts. a, Two sentences scored word by word. Bars above the line are hedons, bars below are dolors; grey words are buffer words and carry no charge. *Good* sits inside the reach of *not*, *wonderful* follows an intensifier, and the second sentence follows a contrast word. **b**, Entries of the signature of each sentence and of the two read as one text, obtained by merging and equal to a single pass over both. **c**, Readouts of the merged signature: totals, the level with its standard error (shipped α and κ for the level), and the cubic arc of net charge over relative position.

position kernel fitted in each corpus puts more weight on the first third of a text than on the last third in SST sentences (difference +0.26 [+0.14, +0.39]), less in tweets (-0.39 [-0.54, -0.24]) and Yelp reviews (-0.12 [-0.15, -0.09]), and is undecided in the other three (Fig. 3c). In the shipped model, fitted on all six corpora, the last third outweighs the first (-0.13 [-0.20, -0.07]).

The third test uses human ratings alone. The SST gives a human rating to every phrase of its parse trees [39], so the rating of a phrase can be regressed on the ratings of its two parts. Across 154,457 phrase pairs the later part gets the larger weight: 0.68 against 0.61 when both parts have at least two words (difference +0.073 [+0.055, +0.089]; Fig. 3b). The later part of an English phrase is often its predicate, so this is a fact about where evaluation sits in syntax as much as about order. Either way the early words do not dominate. Within EmoBank documents, rated intensity does not fall with sentence position either (slope +0.012 [-0.012, +0.032] per document length; 115 documents).

Front-loading is therefore refuted as a general rule. Earlier work on reviews found the same for whole documents, where the closing sentences carry the verdict [43–46], and it fits the recency effects known from memory and from retrospective judgements of experience [47,48].

Neutral words dilute, when they follow the feeling

Dilution fares better (Fig. 4). Holding a text's total charge fixed, rated intensity falls as the number of buffer words rises in 5 of six corpora, with partial rank correlations from -0.27 in EmoBank to -0.08 in SST sentences. Whole film reviews show no effect (+0.01 [-0.02, +0.04]). Held-out agreement peaks at a buffer weight κ between 0.2 and 1 in the same five corpora and at zero for film reviews (Fig. 4c).

Human phrase ratings show the effect without any scorer. Take a phrase with a clear rating and join it to a neutral phrase of m words. When the neutral words come after the rated phrase, the joined phrase keeps 0.78 of the rating for $m = 1$ and 0.62 [0.57, 0.67] for $m \geq 9$. When they come before it, the share kept is 0.75 for $m = 1$ and 0.76 [0.71, 0.80] for $m \geq 9$: no decline (Fig. 4a). The level of these shares is pulled down by regression to the mean, because the rated phrase was selected for a strong rating; the comparison between the two orders, and the trend with m, are not.

So the 2024 manuscript had dilution right and its direction of travel wrong. Trailing neutral words weaken a judgement and leading ones do not, which is recency once more: what a reader meets last counts most, and if that is neutral the feeling fades.

A book at every scale

I scored 356 novels from Project Gutenberg (28.1 million words, median 25 chapters) sentence by sentence and merged the signatures up to chapters and books. Fig. 5a,b shows *Moby-Dick*: 213,523 words carrying 11,790 hd and 3,539 dl, a density of +39 per thousand words, with each of its 129 chapters and twelve spans of its final chapter read from the same pass.

Chapters differ by more than chance. The variance of chapter densities is 5.9 times (median; interquartile range 4.3 to 7.8) what results when a novel's sentences are shuffled across its chapters, and exceeds the shuffled value at $p \leq 0.01$ in 98% of novels (Fig. 5c). One number for a book hides most of what its chapters do.

The cubic arc read from a book's signature captures 41% (median) of the variance of an arc drawn with a moving window one tenth of the book wide; keeping moments up to degree nine would capture 63% (Fig. 5d). The two leading modes of variation among the window arcs of the 356 novels are close to the first two Legendre polynomials (cosine similarity 0.94 and 0.89), the rise or fall and the single turn that earlier studies of story shapes found [24]; the third mode is not close to the cubic (0.10).

Human labels exist for one body of narrative. Alm and colleagues labelled every sentence of 176 fairy tales with an emotion [49]. The cubic arc of each tale's merged signature agrees in sign with the cubic arc of its human labels in 70% of tales (median correlation 0.46; sign test $p = 6 \times 10^{-8}$; Extended Data Fig. 7c). For a tale's overall valence the picture is less favourable. The calibrated score of the merged signature correlates with the mean human label at 0.50 [0.34, 0.63], and VADER averaged over sentences does better (0.63 [0.51, 0.73]). The human target here is itself a mean of sentence labels, which an average of sentence scores mirrors. VADER applied to the whole tale saturates beyond ±0.99 for 73% of tales and falls to 0.39.

A caution applies to every number in this section. Word norms rate words, not events. The final chapter of *Moby-Dick*, in which the ship is destroyed, scores near the book's own level, because *whale*, *sea* and *boat* are pleasant words to raters. Densities of narrative text are best read against the text's own baseline.

Pooling, averaging and chunking

Bounded scores depend on how a long text is cut (Fig. 6a–c). For 60 of the novels I computed a book score as the mean VADER compound score over chunks. The mean across books rises from 0.11 with

one-sentence chunks to 0.76 with whole chapters, and 19% of pairs of books swap order between the two. Two transformer classifiers behave the same way on 300 whole film reviews: the score of the review and the mean score of its sentences differ in sign for 28% of reviews with DistilBERT and 51% with the RoBERTa model, and in both cases the mean of sentence scores predicts the critic's rating better than the score of the whole review does ($r = 0.61$ against 0.53, and 0.52 against 0.23); 97% of the reviews exceed the models' 512-token window and are truncated.

Averaging is the problem, whatever is averaged. If the densities of chunks are averaged, 8% of pairs of books swap order as the chunk size changes. If the chunks' signatures are merged, the book's density is the same for every chunking, to a relative difference below 1×10^{-14}.

How far apart are a pooled value and a mean of parts? For parts with exposures D_p and densities λ_p,

$$\frac{\sum_p D_p \lambda_p}{\sum_p D_p} = \overline{\lambda} + \frac{\operatorname{Cov}(D, \lambda)}{\overline{D}}, \tag{4}$$

an identity familiar from the decomposition of aggregate productivity [50]. The two differ when long parts differ in density from short ones. In novels they rarely do: the median gap between a novel's pooled density and the mean of its chapter densities is 4% of the standard deviation of density between novels, and 3.1% of the 63,190 pairs of novels are ordered differently by the two (Fig. 6d).

A difference between two bodies of text splits in the same spirit. Kitagawa's decomposition [51] writes a difference in pooled density as a part due to differences within strata plus a part due to the mix of strata. Four film critics in the review corpus [40] illustrate it, with the critic's own rating as the stratum (Fig. 6e). Pooled, Berardinelli's and Renshaw's reviews have the same density (difference -0.1 [-1.3, +1.1] per thousand words). Within the same rating Berardinelli's prose is the more negative by -1.2 [-2.5, +0.0], an interval that reaches zero, and his mix of ratings adds back +1.1 [+0.8, +1.6]. The decomposition is bookkeeping, and which part answers a question depends on the question [52]; the point is that a signature kept per stratum makes both parts available, and a pooled bounded score does not.

Discussion

Pending.

Methods

Word charges. labMT 1.0 gives the mean of 50 ratings of happiness on a scale from 1 to 9 for 10,222 frequent English words [2]. The charge of a word is $c = (h - 5)/4$, so that $c = 1$ at the top of the scale, 0 at its neutral point and -1 at the bottom. The VADER list gives the mean of 10 ratings from -4 to 4 for 7,494 words, emoticons and abbreviations [7]. I put it on the labMT scale with one slope through the origin, the ratio of root-mean-square values over the 1,366 shared words, and averaged the two values where both exist. The result has 16,350 entries, of which 8,408 have $|c| \geq 0.25$ and carry charge; the rest are buffer words, as are words outside the list. Positive charge is $h = \max(c, 0)$ hedons and negative charge $d = \max(-c, 0)$ dolors. Charges are stored as integers in thousandths. Adding the NRC-VAD [53] and Warriner [54] norms ($r = 0.88$ and 0.92 with labMT on shared words) raised mean agreement on development data from 0.521 to 0.534; their licences do not allow redistribution, so they are not part of the shipped lexicon and no reported result uses them.

Tokens and operators. Text is lower-cased. Emoji are replaced by their names, using the list distributed with VADER, and 40 emoticons are recognised when they follow a space. Words are runs of letters and apostrophes. A full stop, exclamation mark or question mark ends a sentence; a comma, semicolon or colon ends a clause. Four closed lists of operator words carry no charge themselves: negators (*not*, *never*, forms ending in *n't* and others), intensifiers (*very*, *really*, *so*), diminishers (*slightly*, *somewhat*) and contrast words (*but*, *however*, *yet*, *nevertheless*, *nonetheless*). A negator reaches the next three words or the next punctuation mark, whichever comes first; an intensifier or diminisher reaches the next two words; a contrast word reaches the end of its sentence. Every sentence end clears all operators. The scan is a finite-state transducer and uses string replacement, translation and splitting only.

Signature. For a span of L words with charges h_i, d_i at positions $i = 0, \ldots, L-1$, the signature holds L; the number E of charged words; $S_k^{\pm}$ for $k = 0, \ldots, 3$ (equation 1); the hedons and the dolors inside the reach of each of the four operators; and $\sum_i (h_i - d_i)^2$. The fast implementation computes signatures for a batch of texts with array operations. A reference implementation loops over tokens in integer arithmetic and defines the result; the test suite checks that the two agree, and checks the propositions below for exact equality on random texts.

Proposition 1. Let A and B be texts and let A end at a sentence end. Then $\sigma(AB) = \sigma(A) \oplus \sigma(B)$, where $\oplus$ adds every entry except the position sums and applies equation (2) to those. The operation $\oplus$ is associative, the zero vector is its identity, and for fixed $\sigma(A)$ the map $\sigma(B) \mapsto \sigma(A) \oplus \sigma(B)$ is invertible.

Proof. Operators do not reach across a sentence end, so each word of B has the same charge and the same operator context in AB as in B. Counts and totals therefore add. A word at position j of B is at position $L_A + j$ of AB, and expanding $(L_A + j)^k$ gives equation (2). Write T_ℓ for the linear map that equation (2) applies to the position sums for a shift by ℓ. It is the matrix of the substitution $x \mapsto x + \ell$ on polynomials of degree below four in the monomial basis, so $T_\ell T_m = T_{\ell+m}$ and $T_\ell^{-1} = T_{-\ell}$, which give associativity and invertibility. □

Merging is not commutative: parts can be scored in any order and on any machine, and must be merged in reading order. With prefix signatures $\sigma_j = \sigma(s_1 \cdots s_j)$ over sentences, the signature of sentences i to j is $T_{-L}(\sigma_j - \sigma_{i-1})$ in the position sums, with L the length of the first $i - 1$ sentences. In floating point this difference loses precision for high moments of short spans deep inside long texts; the integer implementation does not.

Proposition 2. Let $\varphi_1, \ldots, \varphi_n$ be linearly independent real sequences on positions $0, 1, 2, \ldots$, and for a real charge sequence c of length L let $\Phi(c) = \big(L, \sum_i c_i \varphi_1(i), \ldots, \sum_i c_i \varphi_n(i)\big)$. There is a function F with $\Phi(cc') = F(\Phi(c), \Phi(c'))$ for all charge sequences c, c' if and only if the span V of the φ_r is closed under the shift $(T\varphi)(i) = \varphi(i+1)$. In that case every φ_r satisfies a linear recurrence with constant coefficients of order at most n, and is a combination of sequences $i^k \rho^i$, with ρ a nonzero complex number, and of sequences that vanish from position n onward.

Proof. If V is closed under the shift, then $T^\ell \varphi_r = \sum_s M_{rs}^{(\ell)} \varphi_s$ for some matrices $M^{(\ell)}$, and $\sum_j c'_j \varphi_r(L + j) = \sum_s M_{rs}^{(L)} \sum_j c'_j \varphi_s(j)$, so $\Phi(cc')$ follows from $\Phi(c)$ and $\Phi(c')$. Conversely, fix $\ell \geq 1$ and let c be ℓ zeros. Take J large enough that the restrictions of the φ_s to positions below J are linearly independent, and let c' range over sequences of length J. By assumption $\sum_j c'_j \varphi_r(\ell + j)$ depends on c' only through the numbers $\sum_j c'_j \varphi_s(j)$. Adding to c' any d with $\sum_j d_j \varphi_s(j) = 0$ for all s therefore leaves it unchanged, so the linear functional $d \mapsto \sum_j d_j \varphi_r(\ell + j)$ vanishes on the common kernel of the n functionals $d \mapsto \sum_j d_j \varphi_s(j)$ and is a linear combination of them. The coefficients are unique, and do not change with J because the equations for J are among those for $J + 1$. Hence $T^\ell \varphi_r \in V$. The shift is then a linear map on the n-dimensional space V and, by the Cayley–Hamilton theorem, is annihilated by its characteristic polynomial p, so every member of V satisfies the recurrence $p(T)\varphi = 0$. Its solutions are spanned by $i^k \rho^i$ for the nonzero roots ρ of p and, for a zero root of multiplicity m, by the sequences supported on positions below m. □

The proposition concerns summaries that are linear in the charges. It is the sequence form of the description of finite-dimensional translation-invariant function spaces [29,30], and of the fact that a function on strings has a finite-dimensional linear representation exactly when its Hankel matrix has finite rank [31,32]. A geometric weight ρ^i gives the one-number case $S_\rho(AB) = S_\rho(A) + \rho^{L_A} S_\rho(B)$. On relative position: a weight $g\big((i + \frac{1}{2})/L\big)$ is available from a summary for every length L only if the span of weights is closed under rescaling of position as well as under shifts. A finite-dimensional space of functions of a real variable with both properties is the space of polynomials up to some degree: closure under shifts gives exponential polynomials [29], rescaling rules out any nonzero exponent λ because the functions $e^{\lambda s x}$, $s > 0$, are linearly independent, and a shift-invariant space of polynomials contains the derivatives of its members.

Readouts. Density is $1000\,(H - D)/L$. Level is equation (3); its standard error is $1000\sqrt{\sum_i (h_i - d_i)^2 - (H - D)^2/L} \,/\, (a + E + \kappa B)$, which treats the charges of a text's words as independent draws. With $x_i = (2i + 1)/L - 1$ and Legendre polynomials P_k, the moments $m_k = \sum_i (h_i - d_i) P_k(x_i)$, $k \leq 3$, are linear combinations of the position sums, and the arc is $f(x) = \sum_k \frac{2k+1}{L} m_k P_k(x)$. The sampled polynomials are orthogonal up to terms of order L^{-2}, so f is the least-squares cubic to that order; over the novels it agrees with a direct fit to 10^{-8}. The calibrated valence is $b_0 + \mathbf{b} \cdot \mathbf{x}/(a + E + \kappa B)$, where $\mathbf{x}$ holds H, D, the moments m_1, m_2, m_3, the negated hedons, the negated dolors, and the net charge that is intensified, diminished, and after a contrast word.

Corpora. Table 1 lists the rated corpora. Official training, development and test portions were kept where they exist. Facebook posts and film reviews were split 40/10/50 by a seeded shuffle. Yelp training and development portions are seeded samples of the official training set.

Fitting and evaluation. Ratings were standardised within each corpus with the mean and standard deviation of its training portion, and corpora were given equal total weight. For each target corpus and each pair (a, κ) on the grid $a \in \{0, 0.5, 1, 2, 4, 8, 16, 32\}$, $\kappa \in \{0, 0.05, 0.1, 0.2, 0.5, 1, 2\}$, the coefficients were fitted by ridge regression on the training portions of the other five corpora (penalty 10^{-3} times the mean diagonal of the weighted Gram matrix; intercept unpenalised). The pair with the highest mean Pearson r on those corpora's development portions was kept, the coefficients were refitted on training plus development portions, and the target's test portion was scored once. The

Table 1 Rated corpora. Numbers of texts in the training, development and test portions.

Corpus	Unit	Human rating	Train / dev / test
EmoBank [36]	sentence, six genres	valence, 1 to 5	8,062 / 1,000 / 1,000
SemEval-2018 [37]	tweet	valence, 0 to 1	1,181 / 449 / 937
Facebook posts [38]	post	valence, 1 to 9	1,158 / 289 / 1,448
SST [39]	sentence	sentiment, 0 to 1	8,544 / 1,101 / 2,210
Film reviews [40]	review	critic's rating	2,002 / 501 / 2,503
Yelp [41]	review	stars, 1 to 5	100,000 / 10,000 / 50,000

shipped constants ($a = 4$, $\kappa = 0.5$) are the same fit on all six corpora. Intervals are 95% percentile intervals from 2,000 bootstrap resamples, drawing source documents for EmoBank and items elsewhere; differences between methods use the same resamples. Comparators: VADER compound score; the hedonometer, the mean labMT value of words outside its neutral band [2]; TextBlob polarity; ridge regression on TF-IDF weights of words and word pairs, trained on the target's own training portion or on the other five corpora; and two transformer classifiers scored as $P(\text{positive}) - P(\text{negative})$ on text truncated to 512 tokens, DistilBERT fine-tuned on SST-2 [12] and a RoBERTa model fine-tuned on tweets [42]. The transformers were run on a seeded sample of 5,000 Yelp test reviews, and for Yelp all methods are compared on that sample.

Refined charges. With exposure $a + L$ and operator multipliers θ from a one-channel fit, the predicted rating of text t is $b_0 + s \sum_i \mu_i c_{w_i} / (a + L_t)$ with $\mu_i = 1 + \theta \cdot (\text{operator flags of word } i)$, which is linear in the vector c of word charges. The refined charges minimise the weighted squared error plus $\lambda \| s(c - c^0) \|^2$, where c^0 holds the norm charges and words that occur in at least 25 training texts start at zero. The penalty $\lambda \in \{0.03, 0.1, 0.3\}$ was chosen on development data of the fitting corpora, and the readout was then refitted with the refined charges. Each step is a ridge regression; the sequence is not a joint optimum.

Position and dilution. For where charge sits, each text of at least ten words contributes $\sum_i |c_i| x_i$ and $\sum_i |c_i|$, and the statistic is the ratio of their sums over texts. Besides the six rated corpora, the units were paragraphs of the film reviews and chapters of 150 of the novels. The position kernel of a corpus is $1 + (b_1 P_1 + b_2 P_2 + b_3 P_3)/b_0$ from a one-channel fit to all its portions with the shipped a and κ; the contrast is its mean over the first third minus its mean over the last third, with intervals from 1,000 resamples. For the treebank, every binary node of the training trees whose phrase and both parts have ratings in the phrase dictionary was used (100% of nodes); ratings were centred at 0.5, the phrase rating was regressed on the two part ratings, and sentences were resampled. For dilution in the treebank, a part counts as rated if its centred rating is at least 0.15 in absolute value and as neutral if at most 0.05; the share kept is the ratio of sums of the signed phrase rating to the absolute part rating. For dilution in the corpora, rated intensity is the absolute deviation of a rating from the corpus median, and the partial rank correlation with the number of buffer words removes the rank of total charge from both by linear regression. For EmoBank documents of eight or more sentences, $|V - 3|$ was regressed on relative sentence position with document means removed, resampling documents.

Novels, fairy tales and critics. From a seeded sample of English fiction in the Project Gutenberg catalogue, 640 texts were fetched from a mirror, and 356 had at least ten chapters of 300 or more words and 20,000 words in all after removal of front matter. Chapters were found by headings and closed with a full stop. The null for chapter differences shuffles a novel's sentences 200 times and keeps the number of sentences in each chapter. Arcs were compared on 100 points; the moving window is 10% of the words. The fairy-tale annotations [49] give two annotators' emotion labels for each sentence; a label of happy or positively surprised was scored +1, neutral 0, and any other emotion −1, and the two were averaged. Critic comparisons use all 5,006 reviews with the four-level rating as stratum; intervals resample reviews within critic.

Throughput. Pending.

Pre-specification. The method, the corpora, the outcomes, the comparators and the decision rules for the hypotheses were written into an analysis plan and committed to the repository (commit `7524c58`) before any evaluation data were scored. Test portions were first scored with the plan as written. Deviations since then are listed in the plan: the grid for a and κ was widened after the first run chose its upper edge, without changing the choice; the tokeniser was rewritten for speed; a bug in the selection of constants for the uncalibrated level was fixed; refined charges, the bag-of-words ridge trained on other corpora, the treebank analyses and the check of error against exposure were added and are exploratory. The comparison of human and machine writing was specified in an addendum committed before any machine text was generated (commit `4b10b12`). A February 2026 version of the method, built on word embeddings, failed its own criterion of matching VADER on GoEmotions; the criterion carried over here is a mean r over the primary corpora at least that of VADER.

Software. Python 3.10, NumPy 2.0, SciPy 1.15, scikit-learn 1.5, PyTorch 2.11 and Transformers 4.51. Figures were drawn with Matplotlib 3.9 and SciencePlots [55].

Use of AI tools. The aims and the two hypotheses come from the author's 2024 manuscript. The code, the analyses and the text of this draft were produced with Claude (Anthropic), an AI assistant working in a coding environment under the author's direction, and were reviewed by AI reviewer agents whose reports are in the repository. The essays in the comparison of human and machine writing were written by Claude models by design. Responsibility for the content rests with the author.

Data availability

All corpora and lexicons are public and are downloaded by `scripts/fetch_data.py`; none is redistributed except the open lexicon derived from labMT (CC BY) and the VADER word list (MIT). The machine-written essays, the result files behind every number and figure, and the analysis plan are in the repository.

Code availability

`https://github.com/ayazkhan27/anonymity`. Every number in the text is generated from the result files by `scripts/make_numbers.py`.

Declarations

Funding, acknowledgements and competing-interest statements are to be completed by the author before submission.

References

[1] Pang, B., Lee, L. & Vaithyanathan, S. Thumbs up? Sentiment classification using machine learning techniques. In *Proceedings of the 2002 Conference on Empirical Methods in Natural Language Processing (EMNLP 2002)*, 79–86 (Association for Computational Linguistics, 2002). URL `https://aclanthology.org/W02-1011/`.

[2] Dodds, P. S., Harris, K. D., Kloumann, I. M., Bliss, C. A. & Danforth, C. M. Temporal patterns of happiness and information in a global social network: Hedonometrics and Twitter. *PLoS ONE* **6**, e26752 (2011). Data Set S1 is the labMT 1.0 word list (doi:10.1371/journal.pone.0026752.s001).

[3] Golder, S. A. & Macy, M. W. Diurnal and seasonal mood vary with work, sleep, and daylength across diverse cultures. *Science* **333**, 1878–1881 (2011).

[4] Proksch, S.-O., Lowe, W., Wäckerle, J. & Soroka, S. Multilingual sentiment analysis: A new approach to measuring conflict in legislative speeches. *Legislative Studies Quarterly* **44**, 97–131 (2019). Published online October 2018.

[5] Ribeiro, F. N., Araújo, M., Gonçalves, P., Gonçalves, M. A. & Benevenuto, F. SentiBench – a benchmark comparison of state-of-the-practice sentiment analysis methods. *EPJ Data Science* **5**, 23 (2016).

[6] Dodds, P. S. & Danforth, C. M. Measuring the happiness of large-scale written expression: Songs, blogs, and presidents. *Journal of Happiness Studies* **11**, 441–456 (2010). Published online 17 July 2009.

[7] Hutto, C. J. & Gilbert, E. VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media* **8**, 216–225 (2014).

[8] Taboada, M., Brooke, J., Tofiloski, M., Voll, K. & Stede, M. Lexicon-based methods for sentiment analysis. *Computational Linguistics* **37**, 267–307 (2011). URL `https://aclanthology.org/J11-2001/`.

[9] Tausczik, Y. R. & Pennebaker, J. W. The psychological meaning of words: LIWC and computerized text analysis methods. *Journal of Language and Social Psychology* **29**, 24–54 (2010).

[10] Devlin, J., Chang, M.-W., Lee, K. & Toutanova, K. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 4171–4186 (Association for Computational Linguistics, Minneapolis, Minnesota, 2019). URL `https://aclanthology.org/N19-1423/`.

[11] Liu, Y. *et al.* RoBERTa: A robustly optimized BERT pretraining approach (2019). `1907.11692`.

[12] Sanh, V., Debut, L., Chaumond, J. & Wolf, T. DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter (2019). Presented at the 5th Workshop on Energy Efficient Machine Learning and Cognitive Computing, NeurIPS 2019, `1910.01108`.

[13] Vaswani, A. *et al.* Attention is all you need. In Guyon, I. *et al.* (eds.) *Advances in Neural Information Processing Systems*, vol. 30, 5998–6008 (Curran Associates, Inc., 2017). URL `https://proceedings.neurips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html`. `1706.03762`.

[14] Simpson, E. H. The interpretation of interaction in contingency tables. *Journal of the Royal Statistical Society: Series B (Methodological)* **13**, 238–241 (1951).

[15] Blyth, C. R. On Simpson's paradox and the sure-thing principle. *Journal of the American Statistical Association* **67**, 364–366 (1972).

[16] Lerman, K. Computational social scientist beware: Simpson's paradox in behavioral data. *Journal of Computational Social Science* **1**, 49–58 (2018).

[17] Thelwall, M., Buckley, K., Paltoglou, G., Cai, D. & Kappas, A. Sentiment strength detection in short informal text. *Journal of the American Society for Information Science and Technology* **61**, 2544–2558 (2010). Erratum: 62(2):419, doi:10.1002/asi.21475.

[18] Cacioppo, J. T. & Berntson, G. G. Relationship between attitudes and evaluative space: A critical review, with emphasis on the separability of positive and negative substrates. *Psychological Bulletin* **115**, 401–423 (1994).

[19] Lowe, W., Benoit, K., Mikhaylov, S. & Laver, M. Scaling policy preferences from coded political texts. *Legislative Studies Quarterly* **36**, 123–155 (2011).

[20] Cormode, G., Shkapenyuk, V., Srivastava, D. & Xu, B. Forward decay: A practical time decay model for streaming systems. In *2009 IEEE 25th International Conference on Data Engineering*, 138–149 (IEEE, 2009).

[21] Pébay, P. Formulas for robust, one-pass parallel computation of covariances and arbitrary-order statistical moments. Tech. Rep. SAND2008-6212, Sandia National Laboratories (2008).

[22] Agarwal, P. K. *et al.* Mergeable summaries. *ACM Transactions on Database Systems* **38**, 1–28 (2013).

[23] Boykin, O., Ritchie, S., O'Connell, I. & Lin, J. Summingbird: A framework for integrating batch and online MapReduce computations. *Proceedings of the VLDB Endowment* **7**, 1441–1451 (2014).

[24] Reagan, A. J., Mitchell, L., Kiley, D., Danforth, C. M. & Dodds, P. S. The emotional arcs of stories are dominated by six basic shapes. *EPJ Data Science* **5**, 31 (2016).

[25] Jockers, M. L. *Syuzhet: Extract Sentiment and Plot Arcs from Text* (2015). URL https://github.com/mjockers/syuzhet. R package. Citation as given by CRAN; current CRAN release is version 1.0.7 (2023), titled "Extracts Sentiment and Sentiment-Derived Plot Arcs from Text".

[26] Bizzoni, Y., Moreira, P., Thomsen, M. R. & Nielbo, K. Sentimental matters – predicting literary quality by sentiment analysis and stylometric features. In *Proceedings of the 13th Workshop on Computational Approaches to Subjectivity, Sentiment, & Social Media Analysis*, 11–18 (Association for Computational Linguistics, Toronto, Canada, 2023). URL https://aclanthology.org/2023.wassa-1.2/.

[27] MacKaye, J. *The Economy of Happiness* (Little, Brown, and Company, Boston, 1906). URL https://archive.org/details/economyhappiness00mackuoft. Uses the "pathon" as the unit of pain intensity (pp. 112–113) and defines the "hedon" as the unit of pleasure intensity (p. 117).

[28] Feldman, F. *Introductory Ethics* (Prentice-Hall, Englewood Cliffs, NJ, 1978). Introduces the "hedon" and the "dolor" as standard units of pleasure and pain (index: Dolor, pp. 24, 34).

[29] Anselone, P. M. & Korevaar, J. Translation invariant subspaces of finite dimension. *Proceedings of the American Mathematical Society* **15**, 747–752 (1964).

[30] Székelyhidi, L. Note on exponential polynomials. *Pacific Journal of Mathematics* **103**, 583–587 (1982).

[31] Schützenberger, M. P. On the definition of a family of automata. *Information and Control* **4**, 245–270 (1961).

[32] Berstel, J. & Reutenauer, C. *Noncommutative Rational Series with Applications* (Cambridge University Press, 2010). Year is the online publication date in the DOI record.

[33] Anderson, N. H. Averaging versus adding as a stimulus-combination rule in impression formation. *Journal of Experimental Psychology* **70**, 394–400 (1965).

[34] Nisbett, R. E., Zukier, H. & Lemley, R. E. The dilution effect: Nondiagnostic information weakens the implications of diagnostic information. *Cognitive Psychology* **13**, 248–277 (1981).

[35] Kiritchenko, S. & Mohammad, S. The effect of negators, modals, and degree adverbs on sentiment composition. In *Proceedings of the 7th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, 43–52 (Association for Computational Linguistics, San Diego, California, 2016). URL https://aclanthology.org/W16-0410/.

[36] Buechel, S. & Hahn, U. EmoBank: Studying the impact of annotation perspective and representation format on dimensional emotion analysis. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, 578–585 (Association for Computational Linguistics, Valencia, Spain, 2017). URL https://aclanthology.org/E17-2092/.

[37] Mohammad, S., Bravo-Marquez, F., Salameh, M. & Kiritchenko, S. SemEval-2018 task 1: Affect in tweets. In *Proceedings of the 12th International Workshop on Semantic Evaluation*, 1–17 (Association for Computational Linguistics, New Orleans, Louisiana, 2018). URL https://aclanthology.org/S18-1001/.

[38] Preoţiuc-Pietro, D. *et al.* Modelling valence and arousal in Facebook posts. In *Proceedings of the 7th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis*, 9–15 (Association for Computational Linguistics, San Diego, California, 2016). URL https://aclanthology.org/W16-0404/.

[39] Socher, R. *et al.* Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, 1631–1642 (Association for Computational Linguistics, Seattle, Washington, USA, 2013). URL https://aclanthology.org/D13-1170/.

[40] Pang, B. & Lee, L. Seeing stars: Exploiting class relationships for sentiment categorization with respect to rating scales. In *Proceedings of the 43rd Annual Meeting of the Association for Computational Linguistics (ACL'05)*, 115–124 (Association for Computational Linguistics, Ann Arbor, Michigan, 2005). URL https://aclanthology.org/P05-1015/.

[41] Zhang, X., Zhao, J. & LeCun, Y. Character-level convolutional networks for text classification. In Cortes, C., Lawrence, N., Lee, D., Sugiyama, M. & Garnett, R. (eds.) *Advances in Neural Information Processing Systems*, vol. 28 (Curran Associates, Inc., 2015). URL https://proceedings.neurips.cc/paper_files/paper/2015/hash/250cf8b51c773f3f8dc8b4be867a9a02-Abstract.html. 1509.01626.

[42] Loureiro, D., Barbieri, F., Neves, L., Espinosa Anke, L. & Camacho-Collados, J. TimeLMs: Diachronic language models from Twitter. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, 251–260 (Association for Computational Linguistics, Dublin, Ireland, 2022). URL https://aclanthology.org/2022.acl-demo.25/.

[43] Pang, B. & Lee, L. A sentimental education: Sentiment analysis using subjectivity summarization based on minimum cuts. In *Proceedings of the 42nd Annual Meeting of the Association for Computational Linguistics (ACL-04)*, 271–278 (Barcelona, Spain, 2004). URL https://aclanthology.org/P04-1035/.

[44] Taboada, M. & Grieve, J. Analyzing appraisal automatically. In *Exploring Attitude and Affect in Text: Theories and Applications. Papers from the 2004 AAAI Spring Symposium* (American Association for Artificial Intelligence, 2004). URL https://cdn.aaai.org/Symposia/Spring/2004/SS-04-07/SS04-07-029.pdf. AAAI Technical Report SS-04-07.

[45] Raychev, V. & Nakov, P. Language-independent sentiment analysis using subjectivity and positional information. In Angelova, G. & Mitkov, R. (eds.) *Proceedings of the International Conference RANLP-2009*, 360–364 (Association for Computational Linguistics, Borovets, Bulgaria, 2009). URL https://aclanthology.org/R09-1065/.

[46] Becker, I. & Aharonson, V. Last but definitely not least: On the role of the last sentence in automatic polarity-classification. In *Proceedings of the ACL 2010 Conference Short Papers*, 331–335 (Association for Computational Linguistics, Uppsala, Sweden, 2010). URL https://aclanthology.org/P10-2061/.

[47] Murdock, B. B., Jr. The serial position effect of free recall. *Journal of Experimental Psychology* **64**, 482–488 (1962).

[48] Fredrickson, B. L. & Kahneman, D. Duration neglect in retrospective evaluations of affective episodes. *Journal of Personality and Social Psychology* **65**, 45–55 (1993).

[49] Alm, C. O., Roth, D. & Sproat, R. Emotions from text: Machine learning for text-based emotion prediction. In *Proceedings of Human Language Technology Conference and Conference on Empirical Methods in Natural Language Processing*, 579–586 (Association for Computational Linguistics, Vancouver, British Columbia, Canada, 2005). URL https://aclanthology.org/H05-1073/.

[50] Olley, G. S. & Pakes, A. The dynamics of productivity in the telecommunications equipment industry. *Econometrica* **64**, 1263–1297 (1996).

[51] Kitagawa, E. M. Components of a difference between two rates. *Journal of the American Statistical Association* **50**, 1168–1194 (1955).

[52] Pearl, J. Comment: Understanding Simpson's paradox. *The American Statistician* **68**, 8–13 (2014).

[53] Mohammad, S. Obtaining reliable human ratings of valence, arousal, and dominance for 20,000 English words. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 174–184 (Association for Computational Linguistics, Melbourne, Australia, 2018). URL https://aclanthology.org/P18-1017/.

[54] Warriner, A. B., Kuperman, V. & Brysbaert, M. Norms of valence, arousal, and dominance for 13,915 English lemmas. *Behavior Research Methods* **45**, 1191–1207 (2013).

[55] Garrett, J. D. garrettj403/SciencePlots. Zenodo (2021). URL https://doi.org/10.5281/zenodo.4106649. Software. The DOI covers all versions; version 1.0.9 alone is doi:10.5281/zenodo.5512926, and the latest release is 2.1.1 (2023).

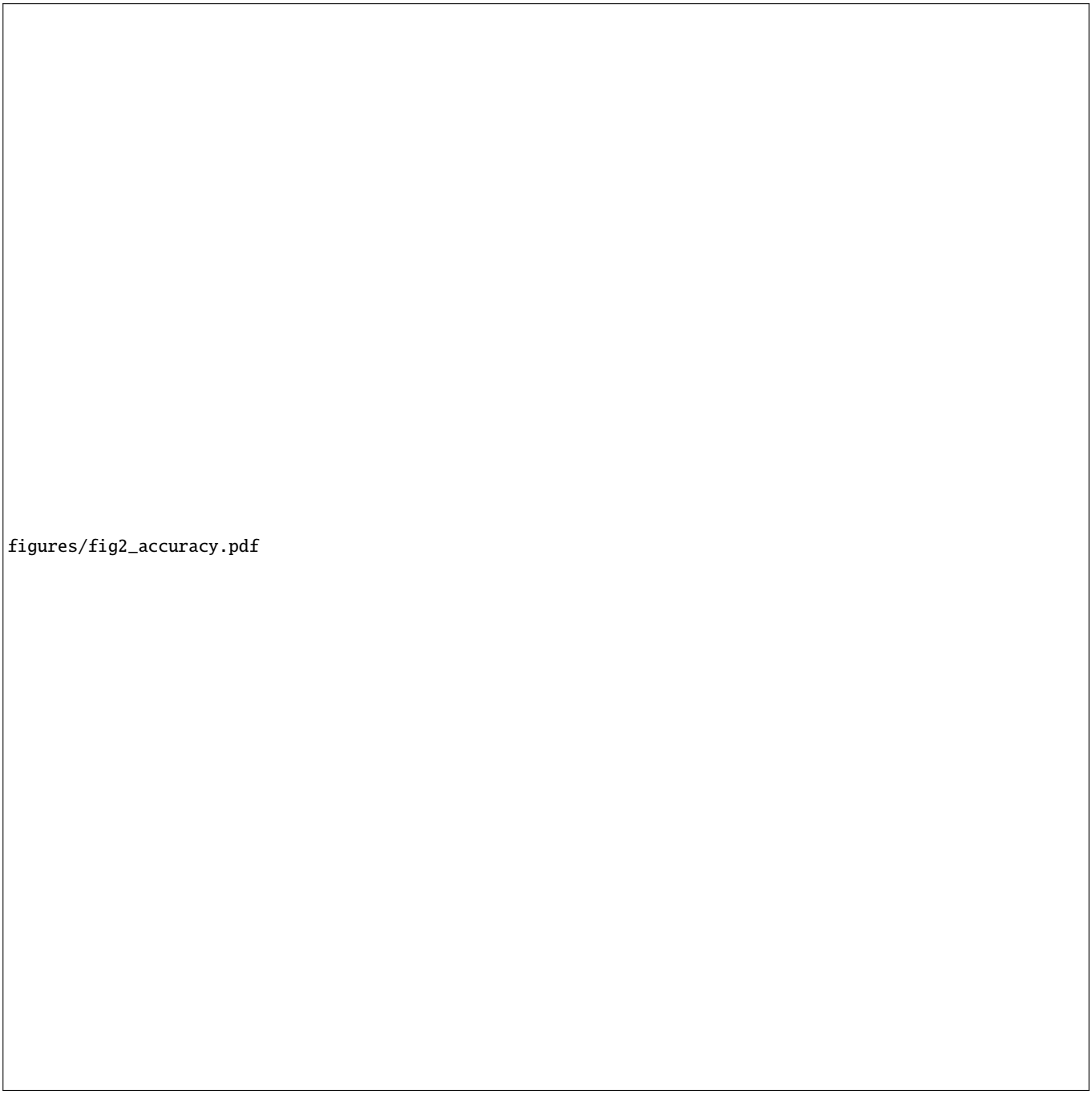


Figure 2 Agreement with human ratings, and cost. a, Pearson r between score and human rating on the held-out test portion of each corpus, with 95% bootstrap intervals. Blue: readouts of the signature, with constants fitted on the other five corpora (filled circle, calibrated valence; open circle, level; square, calibrated valence with refined word charges). Grey: lexicon methods. Green: bag-of-words ridge regression trained on the other five corpora (filled) or on the same corpus (open). Purple: transformer classifiers. **b**, Mean r over the five corpora with continuous human ratings against sentences scored per second on one CPU core. **c**, Loss in r when one part of the signature is removed and the constants are refitted.

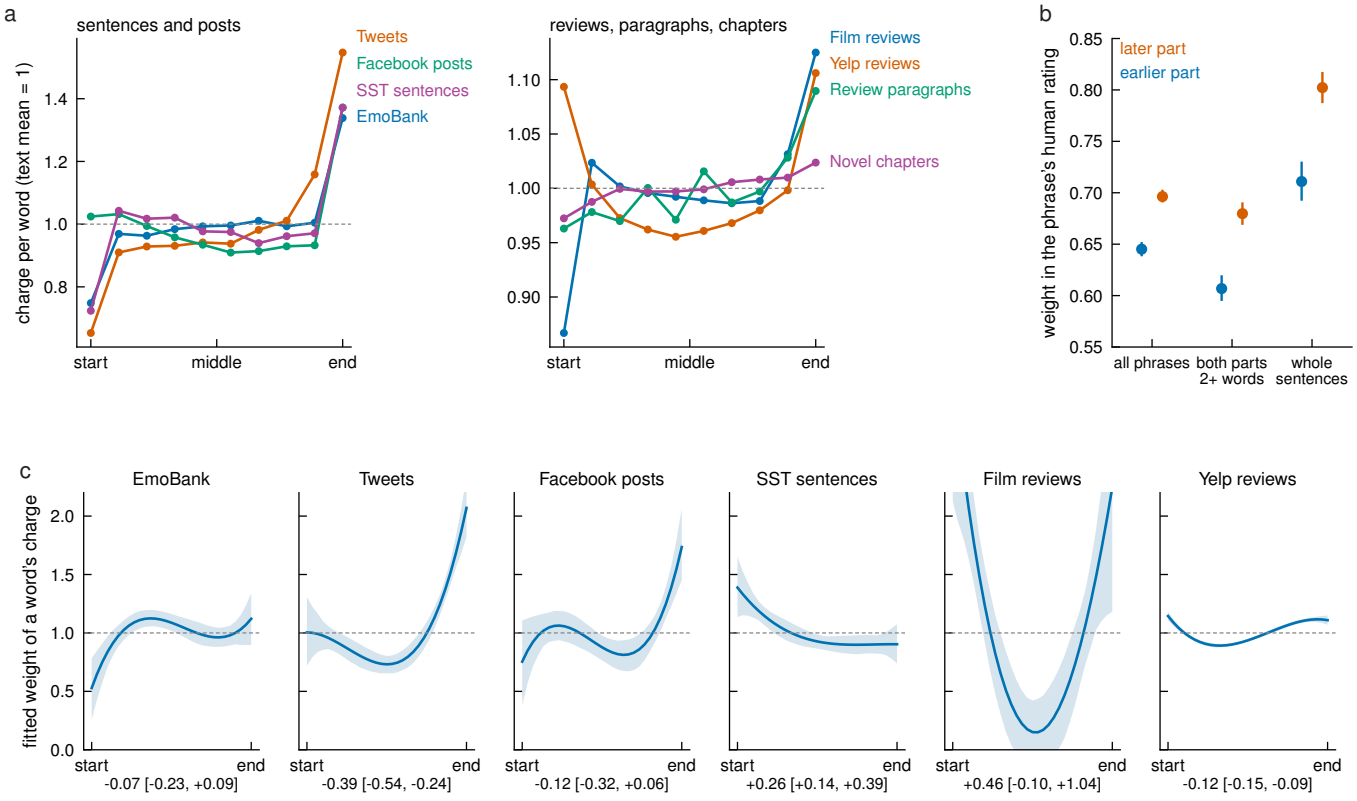


Figure 3 Feeling is not front-loaded. a, Charge per word in each tenth of a text, relative to the text's mean, for texts of ten or more words. **b**, Weights of the earlier and the later part in a regression of a phrase's human rating on the human ratings of its two parts, for binary nodes of the Stanford Sentiment Treebank (95% intervals, sentences resampled). **c**, Position kernel fitted against human ratings in each corpus: the weight given to a word's charge as a function of its relative position, with 95% bands. Below each panel, the mean weight over the first third minus the mean over the last third. The kernel for whole film reviews is poorly determined.

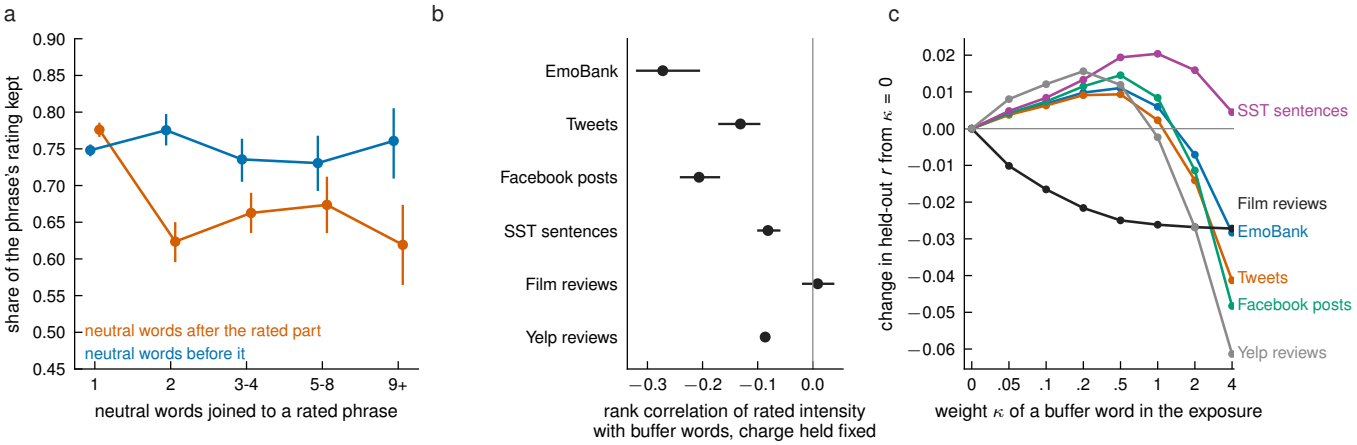


Figure 4 Neutral words dilute a rating when they follow it. a, Human phrase ratings from the Stanford Sentiment Treebank. A phrase with a clear rating is joined to a neutral phrase; the vertical axis is the share of the first phrase's rating that the joined phrase keeps, by the number of neutral words and their side (95% intervals). **b**, Partial rank correlation between rated intensity and the number of buffer words in a text, with total charge held fixed. **c**, Change in held-out r as the weight κ of a buffer word in the exposure moves away from zero, with coefficients refitted at each value.

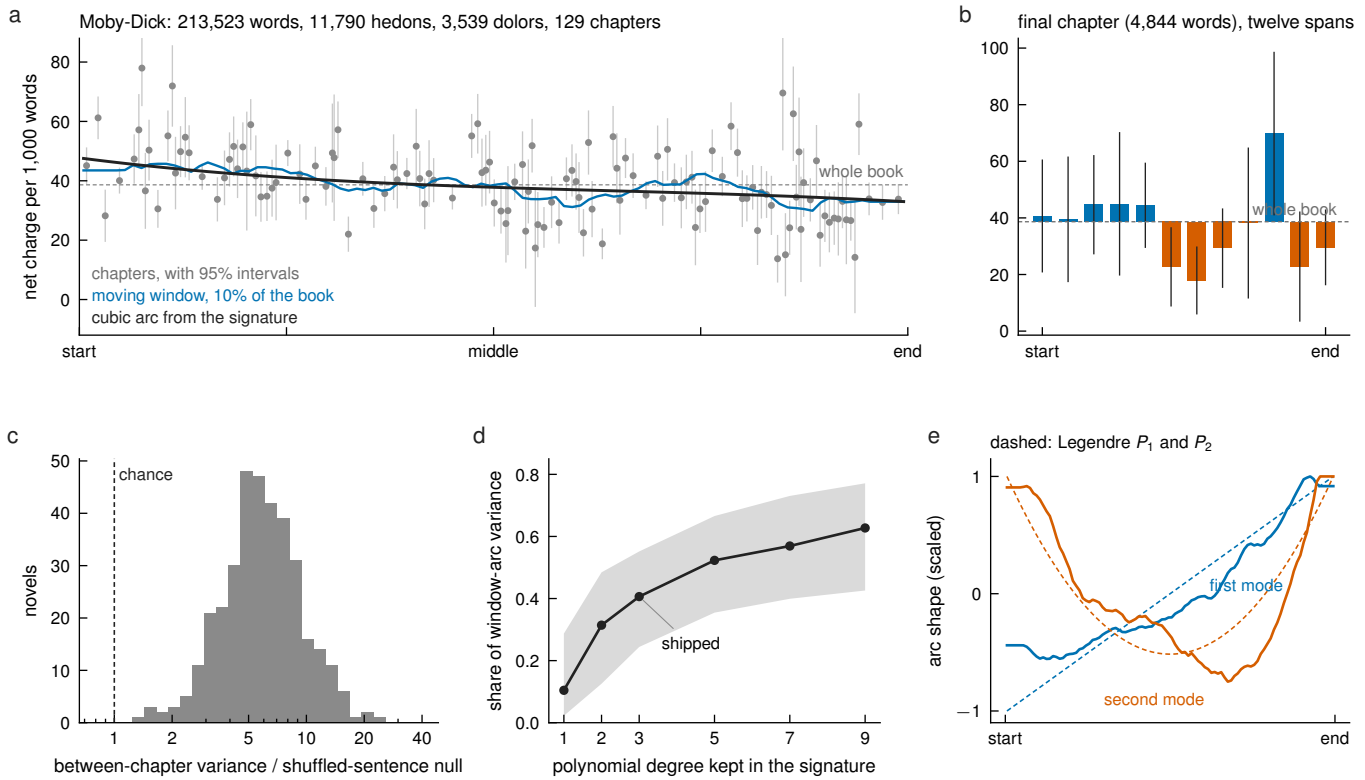


Figure 5 A novel at three scales from one pass. a, *Moby-Dick*: density of each chapter with 95% intervals from the word-level standard error, the cubic arc read from the book's signature, and a moving-window arc for comparison. **b**, The final chapter cut into twelve runs of sentences, each read from two rows of the prefix table; bars show the difference from the whole book's density. **c**, Between-chapter variance of density relative to its value when sentences are shuffled across chapters, for 356 novels. **d**, Share of the variance of the moving-window arc captured by the polynomial arc, by the highest degree kept (median and interquartile range). **e**, The two leading modes of variation among the novels' window arcs, with the first two Legendre polynomials.

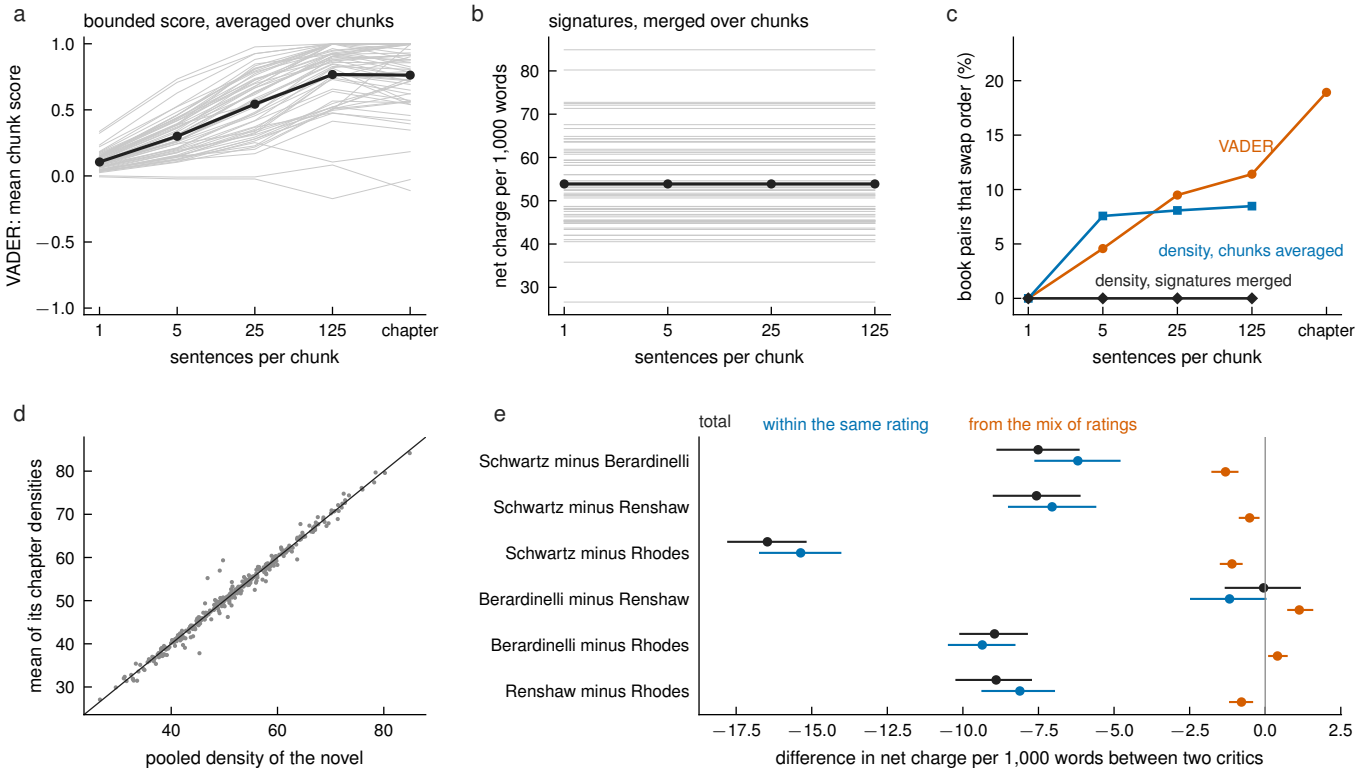


Figure 6 Averaging depends on the cut; merging does not. a, Book score for 60 novels computed as the mean VADER compound score over chunks of a given number of sentences (grey, each novel; black, mean). **b**, Density of the same novels from signatures of the same chunks, merged. **c**, Share of pairs of novels whose order changes relative to one-sentence chunks. **d**, Pooled density of each of 356 novels against the mean of its chapter densities (per 1,000 words). **e**, Differences in density between four film critics, split into the part within the same rating and the part due to the mix of ratings (95% intervals).

Extended data

Table 2 Extended Data Table 1. Pearson r with human ratings on held-out test portions, with 95% bootstrap intervals. The mean is over the five corpora with continuous human ratings (all but Yelp). Signature constants are fitted on the other five corpora unless stated.

	EmoBank	Tweets	Facebook	SST	Film reviews	Yelp	Mean
n (test portion)	1,000	937	1,448	2,210	2,503	5,000	
Signature, calibrated	0.48 [0.43, 0.54]	0.70 [0.67, 0.73]	0.62 [0.58, 0.65]	0.44 [0.40, 0.47]	0.29 [0.26, 0.33]	0.61 [0.59, 0.62]	0.51
Signature, level only	0.44 [0.39, 0.49]	0.71 [0.67, 0.74]	0.57 [0.54, 0.60]	0.42 [0.39, 0.46]	0.26 [0.22, 0.29]	0.54 [0.52, 0.56]	0.48
Signature, constants from the same corpus	0.48 [0.43, 0.55]	0.73 [0.70, 0.76]	0.62 [0.58, 0.65]	0.46 [0.43, 0.49]	0.40 [0.37, 0.44]	0.63 [0.61, 0.64]	0.54
Signature, refined charges	0.41 [0.36, 0.46]	0.71 [0.68, 0.74]	0.58 [0.55, 0.61]	0.55 [0.52, 0.58]	0.49 [0.47, 0.52]	0.64 [0.62, 0.65]	0.55
VADER	0.46 [0.41, 0.50]	0.71 [0.68, 0.75]	0.58 [0.54, 0.62]	0.45 [0.42, 0.48]	0.25 [0.21, 0.29]	0.55 [0.54, 0.57]	0.49
Hedonometer	0.44 [0.40, 0.48]	0.59 [0.55, 0.63]	0.49 [0.45, 0.52]	0.34 [0.30, 0.37]	0.24 [0.20, 0.28]	0.54 [0.52, 0.56]	0.42
TextBlob	0.31 [0.24, 0.36]	0.51 [0.46, 0.56]	0.53 [0.49, 0.57]	0.41 [0.37, 0.44]	0.46 [0.43, 0.49]	0.60 [0.58, 0.61]	0.44
Bag-of-words ridge, other corpora	0.32 [0.27, 0.40]	0.64 [0.60, 0.67]	0.58 [0.54, 0.61]	0.56 [0.53, 0.58]	0.64 [0.61, 0.66]	0.67 [0.65, 0.68]	0.55
Bag-of-words ridge, same corpus	0.51 [0.47, 0.55]	0.59 [0.55, 0.63]	0.53 [0.50, 0.57]	0.66 [0.64, 0.68]	0.77 [0.75, 0.78]	0.83 [0.83, 0.84]	0.61
DistilBERT (SST-2)	0.50 [0.46, 0.53]	0.60 [0.56, 0.64]	0.54 [0.50, 0.57]	0.78 [0.76, 0.79]	0.56 [0.53, 0.58]	0.75 [0.73, 0.76]	0.59
Twitter-RoBERTa	0.73 [0.70, 0.76]	0.81 [0.78, 0.83]	0.78 [0.76, 0.80]	0.77 [0.75, 0.79]	0.31 [0.27, 0.34]	0.76 [0.75, 0.77]	0.68

Table 3 Extended Data Table 2. Charge-weighted mean position of words, from -1 (first word) to $+1$ (last word), for texts of ten or more words, with 95% intervals; and the same for hedons and dolors separately.

Unit	Texts	All charge	Hedons	Dolors
EmoBank	5,975	+0.060 [+0.052, +0.068]	+0.051	+0.107
SemEval-2018 tweets	2,127	+0.105 [+0.096, +0.114]	+0.090	+0.156
Facebook posts	1,698	+0.021 [+0.012, +0.031]	+0.018	+0.036
SST sentences	9,358	+0.050 [+0.046, +0.055]	+0.042	+0.087
Film reviews	5,006	+0.023 [+0.022, +0.025]	+0.024	+0.022
Yelp reviews	157,452	+0.002 [+0.001, +0.002]	-0.001	+0.023
Film review paragraphs	22,514	+0.017 [+0.016, +0.019]	+0.011	+0.041
Novel chapters	2,583	+0.007 [+0.005, +0.009]	+0.005	+0.015

Table 4 Extended Data Table 3. Coefficients of the shipped calibrated readout ($a = 4$, $\kappa = 0.5$), fitted on six corpora. The readout predicts a human valence rating in standard-deviation units.

Signature entry (per unit exposure)	Coefficient
intercept	-0.356
hedons	+5.850
dolors	-9.554
net charge $\times P_1$	+0.880
net charge $\times P_2$	+1.329
net charge $\times P_3$	+1.259
negated hedons	-15.196
negated dolors	+9.643
intensified net charge	+7.817
diminished net charge	-2.533
net charge after contrast	-0.470

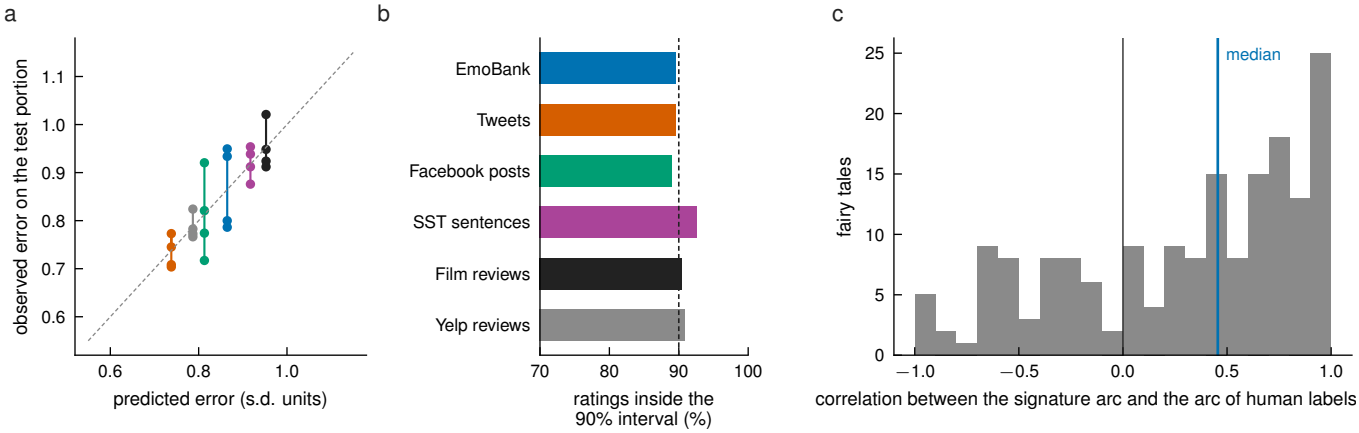


Figure 7 Extended Data Fig. 1. Error against exposure, and arcs of fairy tales. a, Error of the calibrated score against human ratings on each test portion, in quartiles of exposure, against the error predicted by a variance model with a term that shrinks with exposure; the fitted model kept no such term, and observed error does not fall with exposure. **b**, Share of ratings inside the 90% interval implied by the fitted variance. **c**, Correlation between the cubic arc of a tale's merged signature and the cubic arc of its sentence labels from two human annotators, for 176 fairy tales.